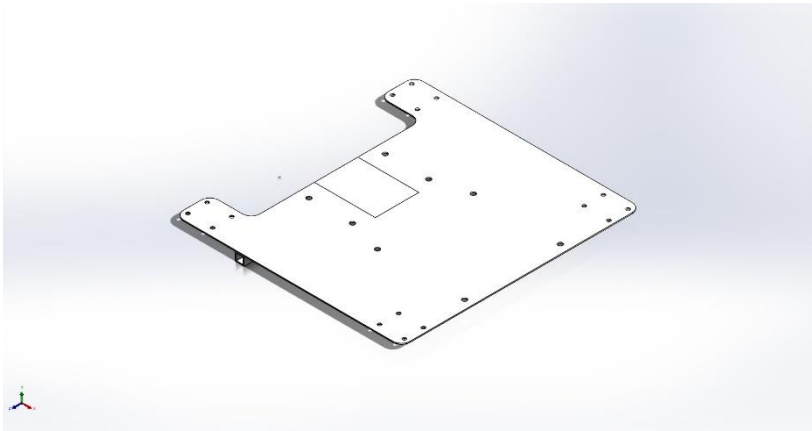




Description
No Data

Simulation of Base Plate

Date: Monday, December 8, 2025
Designer: Solidworks
Study name: Static 1
Analysis type: Static

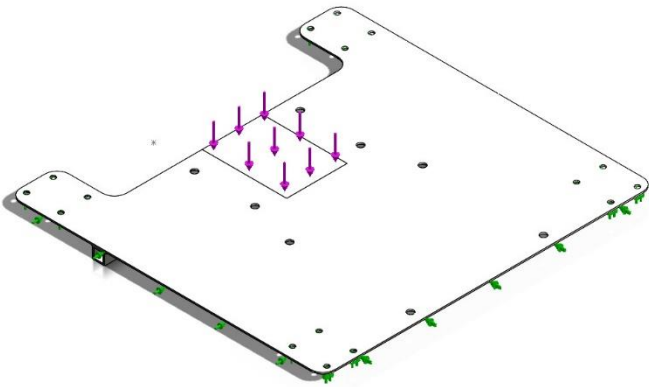
Table of Contents

Description	1
Assumptions.....	2
Model Information	2
Study Properties.....	3
Units	3
Material Properties	4
Loads and Fixtures	5
Connector Definitions	6
Interaction Information	6
Mesh information.....	7
Sensor Details.....	8
Resultant Forces	9
Beams.....	9
Study Results	10
Conclusion.....	13



Assumptions

Model Information



Model name: Base Plate
Current Configuration: Default<As Machined>

Solid Bodies			
Document Name and Reference	Treated As	Volumetric Properties	Document Path/Date Modified
Split Line1	Solid Body	Mass:1.74182 kg Volume:0.000220483 m^3 Density:7,900 kg/m^3 Weight:17.0698 N	C:\Users\masonbuehler\Downloads\CAD-20251208T223626Z-1-001\CAD\Tilty Loader\Base Plate.SLDPRT Dec 8 17:25:41 2025



Study Properties

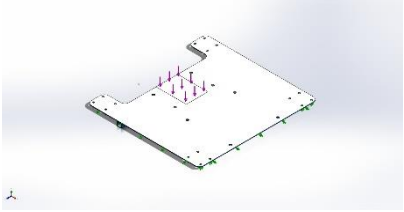
Study name	Static 1
Analysis type	Static
Mesh type	Solid Mesh
Thermal Effect:	On
Thermal option	Include temperature loads
Zero strain temperature	298 Kelvin
Include fluid pressure effects from SOLIDWORKS Flow Simulation	Off
Solver type	Automatic
Inplane Effect:	Off
Soft Spring:	Off
Inertial Relief:	Off
Incompatible bonding options	Automatic
Contact penalty stiffness scale factor	1
Large displacement	Off
Compute free body forces	On
Friction	Off
Use Adaptive Method:	Off
Result folder	SOLIDWORKS document (C:\Users\masonbuehler\Downloads\CAD-20251208T223626Z-1-001\CAD\Tilty Loader)

Units

Unit system:	SI (MKS)
Length/Displacement	mm
Temperature	Kelvin
Angular velocity	Rad/sec
Pressure/Stress	N/m ²

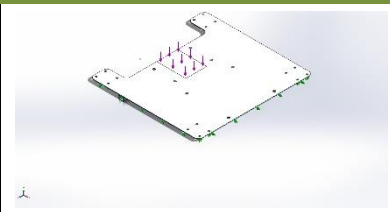


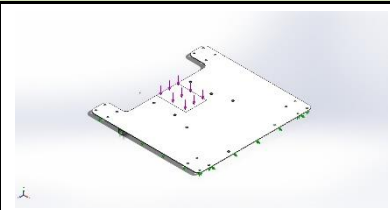
Material Properties

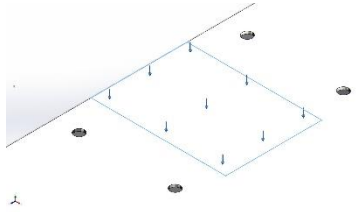
Model Reference	Properties	Components
	Name: AISI 1020 Model type: Linear Elastic Isotropic Default failure criterion: Unknown Yield strength: 3.51571e+08 N/m ² Tensile strength: 4.20507e+08 N/m ² Elastic modulus: 2e+11 N/m ² Poisson's ratio: 0.29 Mass density: 7,900 kg/m ³ Shear modulus: 7.7e+10 N/m ² Thermal expansion coefficient: 1.5e-05 /Kelvin	SolidBody 7(Split Line1)(Base Plate)
Curve Data:N/A		



Loads and Fixtures

Fixture name	Fixture Image	Fixture Details		
Roller/Slider-1		Entities: 2 face(s) Type: Roller/Slider		
Resultant Forces				
Components	X	Y	Z	Resultant
Reaction force(N)	0.00306368	0	0.00117493	0.00328125
Reaction Moment(N.m)	0	0	0	0

Roller/Slider-2		Entities: 16 face(s) Type: Roller/Slider		
Resultant Forces				
Components	X	Y	Z	Resultant
Reaction force(N)	0	222.391	0	222.391
Reaction Moment(N.m)	0	0	0	0

Load name	Load Image	Load Details
Force-1		Entities: 1 face(s) Type: Apply normal force Value: 222.4 N



Connector Definitions

No Data

Interaction Information

No Data



Mesh information

Mesh type	Solid Mesh
Mesher Used:	Blended curvature-based mesh
Jacobian points for High quality mesh	16 Points
Maximum element size	0.792133 in
Minimum element size	0.0765367 in
Mesh Quality	High

Mesh information - Details

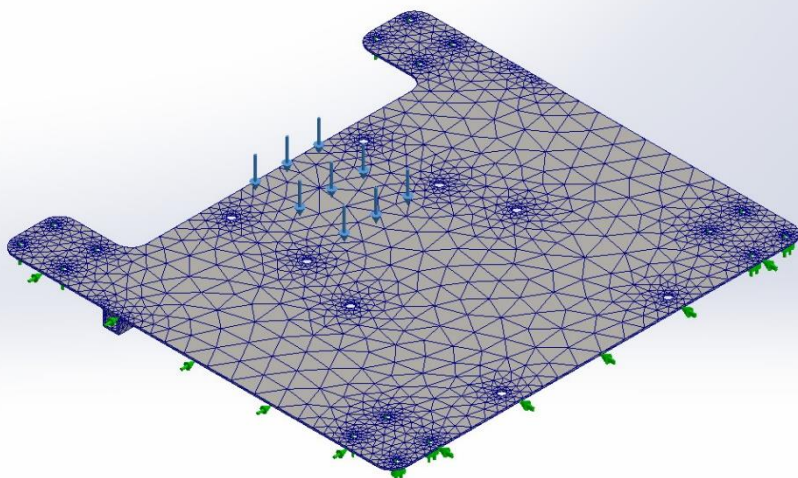
Total Nodes	60753
Total Elements	29666
Maximum Aspect Ratio	58.66
% of elements with Aspect Ratio < 3	26
Percentage of elements with Aspect Ratio > 10	16.8
Percentage of distorted elements	0
Time to complete mesh(hh:mm:ss):	00:00:14
Computer name:	ENG401700

Mesh Quality Plots

Name	Type	Min	Max
Quality1	Mesh	-	-



Model name: Base Plate
Study name: Static 1(-Default<As Machined>-)
Plot type: Mesh Quality1



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Base Plate-Static 1-Quality-Quality1

Sensor Details

No Data



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Analyzed with SOLIDWORKS Simulation

Simulation of Base Plate

Resultant Forces

Reaction forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	0.00306368	222.391	0.00117493	222.391

Reaction Moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	0

Free body forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	-0.630619	-0.832804	0.221476	1.06785

Free body moments

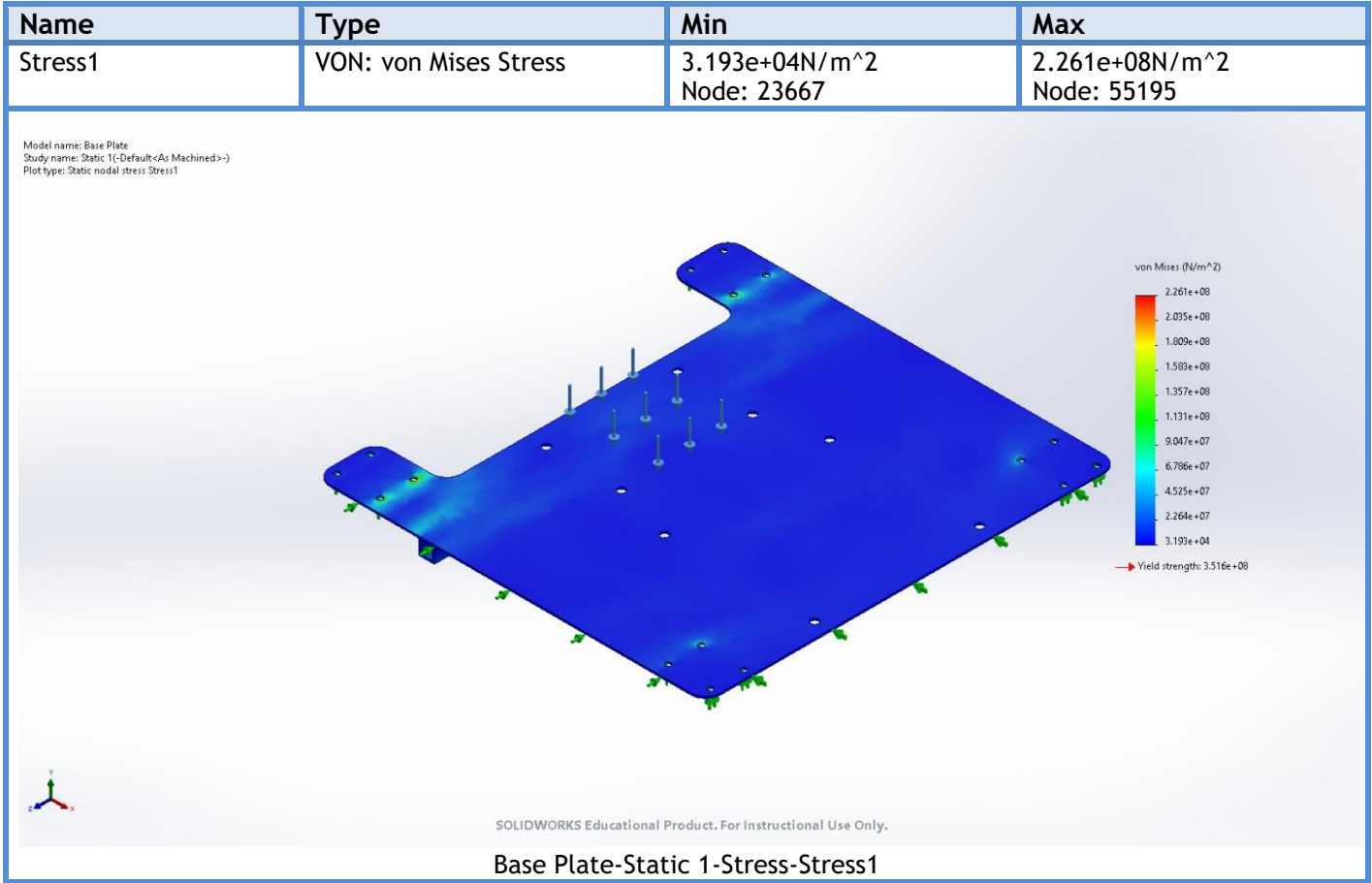
Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	1e-33

Beams

No Data



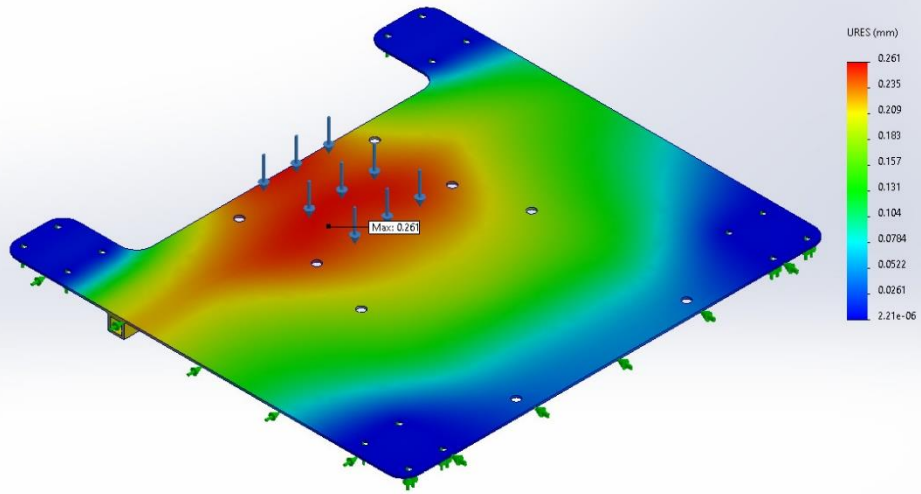
Study Results



Name	Type	Min	Max
Displacement1	URES: Resultant Displacement	2.21e-06mm Node: 20184	0.261mm Node: 47410



Model name: Base Plate
 Study name: Static 1(-Default<As Machined>-)
 Plot type: Static displacement Displacement1

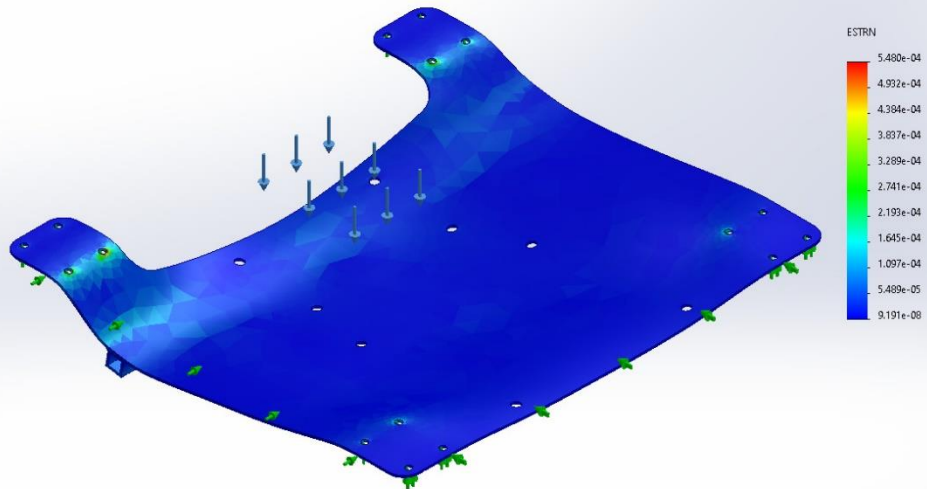


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Base Plate-Static 1-Displacement-Displacement1

Name	Type	Min	Max
Strain1	ESTRN: Equivalent Strain	9.191e-08 Element: 21780	5.480e-04 Element: 13653

Model name: Base Plate
 Study name: Static 1(-Default<As Machined>-)
 Plot type: Static strain Strain1
 Deformation scale: 133.764

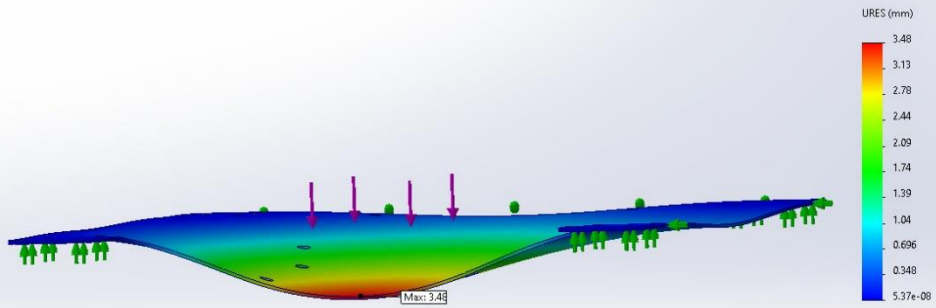


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Base Plate-Static 1-Strain-Strain1



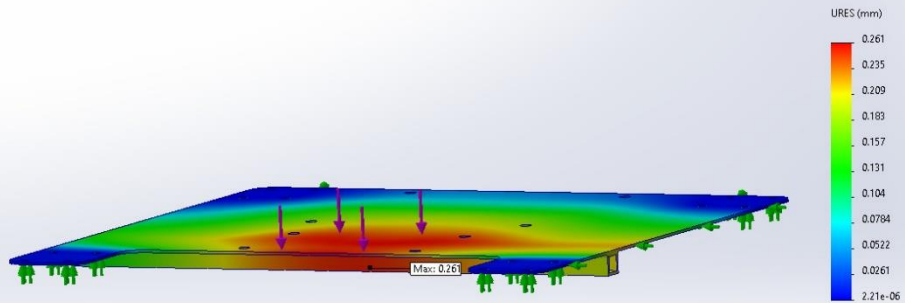
Model name: Base Plate
Study name: Static 1(-Default<As Machined>-)
Plot type: Static displacement Displacement1
Deformation scale: 10.0000



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Image-1

Model name: Base Plate
Study name: Static 1(-Default<As Machined>-)
Plot type: Static displacement Displacement1



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Image-2



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Analyzed with SOLIDWORKS Simulation

Simulation of Base Plate

Conclusion

